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Monetary Tightening and U.S. Bank Fragility in 2023: Mark-to-Market Losses and Uninsured Depositor Runs?

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1 Big picture

- **Question.** How did the 2022–2023 monetary tightening episode affect U.S. bank asset values, and how did mark-to-market losses interact with uninsured deposits to create bank fragility?
- **Main measurement.** The authors mark long-duration bank assets to market using tradable fixed-income and real-estate-linked price indexes.
- **Headline result.** The market value of U.S. banking-system assets declined by roughly 10% on average, with aggregate losses around \$2 trillion relative to book value.
- **SVB interpretation.** SVB was not uniquely bad by asset losses; its distinctive feature was extreme uninsured leverage.
- **Run mechanism.** A bank can survive in a no-run equilibrium because it retains deposit-franchise value. If uninsured depositors run, the bank loses franchise value and may become insolvent.
- **Economic takeaway.** Interest-rate risk and uninsured leverage are complements: hidden duration losses become dangerous when the liability side is runnable.

2 Data and measurement

2.1 Balance-sheet data and mark-to-market losses

The paper uses call-report data for FDIC-insured depository institutions, with Q1 2022 as the pre-tightening reference point. Assets are grouped into RMBS, Treasuries and other securities, residential mortgages, other loans, and cash or non-marked assets. The marked asset categories cover roughly 72% of the aggregate bank asset base.

Interpretation: The paper constructs a market-value balance sheet from regulatory book balance sheets.

2.2 Uninsured leverage and stability measures

Uninsured leverage is uninsured debt divided by assets. The paper constructs uninsured deposit coverage, insured deposit coverage, capital ratio after withdrawals, and extreme insolvency measures.

Interpretation: Asset losses alone do not determine fragility; the bank also needs a runnable liability structure.

3 Models and empirical specifications

3.1 Mark-to-market loss calculation

$$Loss = \sum_t RMBS\ multiplier \times (RMBS_t + Mortgage_t) \times \Delta TreasuryPrice_t + (Treasury\ and\ Other\ Securities\ and\ Loans)_t \times \Delta$$

Interpretation: The loss calculation is a duration-weighted markdown of assets using observed market price declines during monetary tightening.

3.2 No-run and run equity conditions

Let e_b be book equity, c the short-asset share, l_i insured leverage, l_u uninsured leverage, $\Delta a(r_f)$ the asset gain or loss from the rate change, and $\Delta f(r_f)$ the deposit-franchise value.

$$\begin{aligned}\Delta a(r_f) &= \frac{r_0 - r_f}{r_f}, & \Delta f(r_f) &= \frac{r_f - \mu(r_f)}{r_f}. \\ e^{no\ run}(r_f) &= e_b + (1 - c)\Delta a(r_f) + (l_i + l_u)\Delta f(r_f) \geq 0. \\ e^{run}(r_f) &= e_b + (1 - c)\Delta a(r_f) + (l_i + (1 - s)l_u)\Delta f(r_f) < 0. \\ s^* &= \frac{e_b + (1 - c)\Delta a(r_f) + (l_i + l_u)\Delta f(r_f)}{l_u \Delta f(r_f)}.\end{aligned}$$

Interpretation: A run destroys the franchise value of withdrawn uninsured deposits. Higher uninsured leverage lowers the threshold share of awake depositors required for a run equilibrium.

3.3 Empirical solvency diagnostics

$$\begin{aligned}IDCR(s) &= \frac{MTM\ Assets - s \times Uninsured\ Deposits - Insured\ Deposits}{Insured\ Deposits}. \\ Capital\ Ratio(s) &= \frac{Book\ Equity - s \times Uninsured\ Deposits \times \left(\frac{Book\ Asset}{MTM\ Asset} - 1\right)}{Book\ Asset}. \\ Extreme\ Insolvency : &\quad MTM\ Assets < Non-Equity\ Liabilities.\end{aligned}$$

Interpretation: These measures translate the model into bank-level stress tests under different uninsured withdrawal shares.

4 Identification and empirical design

The estimates use public market price changes for comparable long-duration assets and apply them to bank balance-sheet categories. The paper does not estimate depositor run probabilities; it identifies banks for which run equilibria are feasible under specified uninsured withdrawal shares.

5 Tables and figures: expanded interpretations

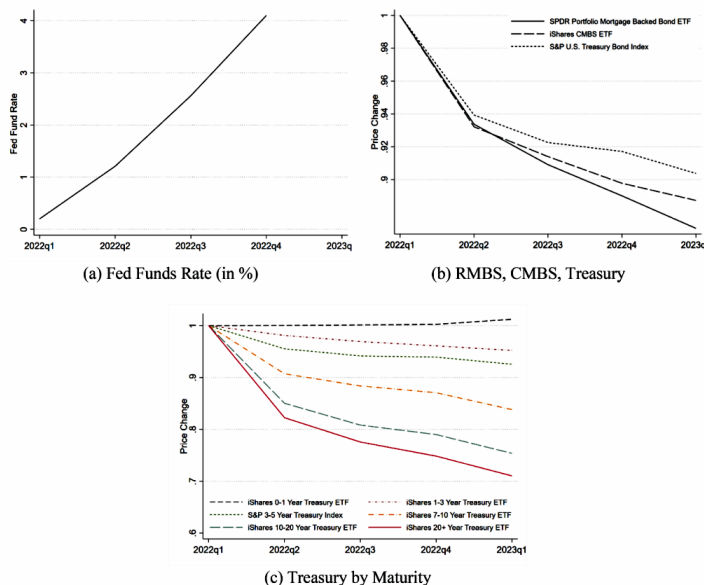
5.1 Figure 1: Fed Tightening and Asset Prices + Table 1: Mark-to-Market Statistics by Bank Size

Read:

- Figure 1A shows the Fed funds rate rising from about 0.08% to about 4.57%.
- Figure 1B shows RMBS, CMBS, and Treasury-linked securities falling in price as rates rise.
- Figure 1C shows longer maturities experiencing larger Treasury price declines.
- Table 1 reports aggregate bank asset losses of about \$2.2 trillion, with loss/assets of 9.2% for all banks, 10.0% for large non-GSIBs, and 7.4% for GSIBs.
- Uninsured deposits divided by marked-to-market assets average 24.2% for all banks, 35.8% for large non-GSIBs, and 37.1% for GSIBs.

Detailed interpretation: Figure 1 is the rate-shock background; Table 1 shows how that shock translates into bank balance-sheet losses and runnable funding exposure. GSIBs have the largest aggregate losses, while large non-GSIBs have the highest loss/assets ratio.

Economic message: Duration losses and uninsured leverage must be evaluated jointly.



(a) Figure 1: Fed Tightening and Asset Prices

Table 1
Mark-to-Market Statistics by Bank Size.

	(1) All Banks	(2) Small (0, 1.384B)	(3) Large (non-GSIB) [1.384B,)	(4) GSIB
Aggregate Loss	2.2T	146.0B	623.2B	1.4T
Bank-Level Loss	28.6M (6.7B)	22.3M (39.1 M)	306.7M (2.3B)	9.9B (62.2B)
Share RMBS	13.2 (19.2)	11.4 (18.5)	22.2 (19.8)	42.6 (34.0)
Share Treasury and Other	15.5 (35.1)	17.0 (37.5)	10.5 (14.6)	7.8 (17.3)
Share Residential Mortgage	19.9 (33.4)	19.8 (35.3)	20.0 (19.5)	24.6 (32.4)
Share Other Loan	32.8 (32.7)	32.7 (34.3)	34.4 (21.5)	7.3 (30.5)
Loss/Asset	9.2 (4.7)	9.1 (4.8)	10.0 (4.4)	7.4 (5.6)
Uninsured Deposit/ MM Asset	24.2 (14.1)	22.7 (12.6)	35.8 (16.0)	37.1 (19.4)
Number of Banks	4844	4090	710	44

This table presents the descriptive statistics of our key metrics after marking to market the asset values for each FDIC-insured depository institution in the

(b) Table 1: Mark-to-Market Statistics by Bank Size

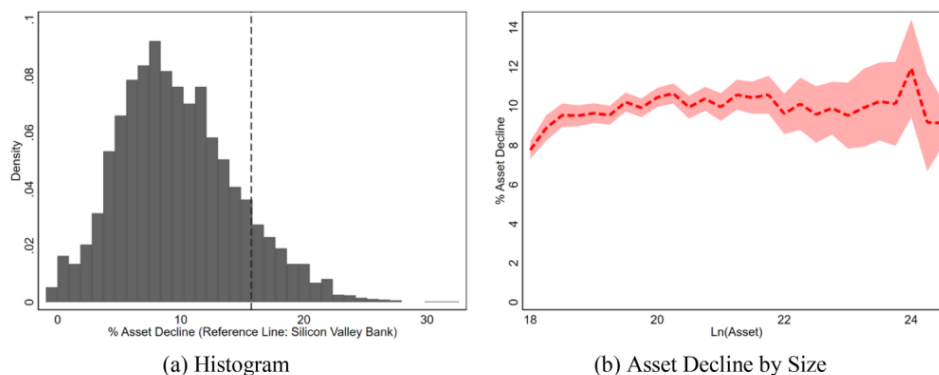
5.2 Figure 2: Distribution of Change in Asset Value (“Marking to Market”)

Read:

- The median asset-value decline is about 9%.
- The 5th, 25th, median, 75th, and 95th percentiles are roughly 4%, 6%, 9%, 13%, and 19%.
- SVB’s asset-value decline is about 15.7%, around the 89th percentile.
- Panel B shows that losses are broad across bank sizes.

Detailed interpretation: SVB is not uniquely fragile by asset losses alone. The figure separates the asset-loss channel from the uninsured-liability channel.

Economic message: Mark-to-market losses are widespread; fragility depends on which banks combine those losses with runnable uninsured liabilities.



of Change in Asset Value ("Marking to Market").

istograms (density) of the percentage of a bank's asset value decline when assets are marked to market according to market Panel a), as well as bank asset value decline by bank size (Panel b). We describe the steps to calculate the mark-to-market ice line in Panel (a) indicates SVB's asset value decline. SVB's asset value declines by 15.7 %, or \$34 billion, after their a: line is at 89th percentile. The 5th, 25th, median, 75th, and 95th percentiles in Panel (a) are 4 %, 6 %, 9 %, 13 %, and 19 % asset value in log terms. The size distribution of the U.S. banking industry has a fat left tail, meaning that there are ma banks' asset sizes range from \$58.9 billion to \$3.5 trillion, while the bottom 10 percentiles have asset values less than \$68 m : about \$66 million, \$485 million, \$3.6 billion, and \$26 billion, respectively. The decline at the right end starts around lo: llion. *Data sources:* Bank call reports in Q1 2022 and various ETF and indices price data as described in the main text.

Figure 2: Distribution of Change in Asset Value ("Marking to Market")

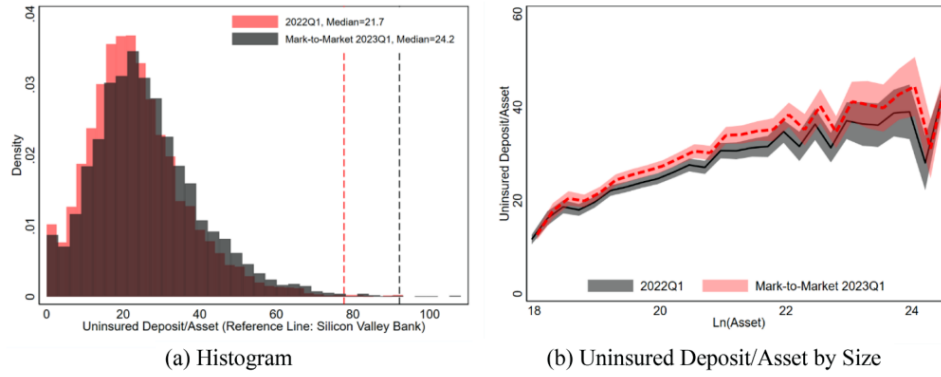
5.3 Figure 3: Distribution of Uninsured Deposit to Asset Ratio (With and Without "Marking to Market")

Read:

- SVB is near the extreme right tail: about 78% uninsured deposits/book assets and about 92% uninsured deposits/marked-to-market assets.
- Marking assets down raises uninsured deposits as a share of assets because the denominator falls.
- Panel B shows uninsured-deposit ratios by bank size.

Detailed interpretation: The figure explains why SVB's failure was not just about losses. The liability side magnified those losses.

Economic message: Accounting values can understate runnable leverage.



of Uninsured Deposit to Asset Ratio (With and Without “Marking to Market”). Histograms (density) of uninsured deposit to asset ratios calculated based on Q1 2022 balance sheets and mark-to-market valuations according to the method described in the main text (Panel a), as well as uninsured deposit ratio against bank size (Panel b). The sample includes all insured depositors to 70% based on the Q1 2022 balance sheet which is less than \$100 billion in assets.

Figure 3: Distribution of Uninsured Deposit to Asset Ratio

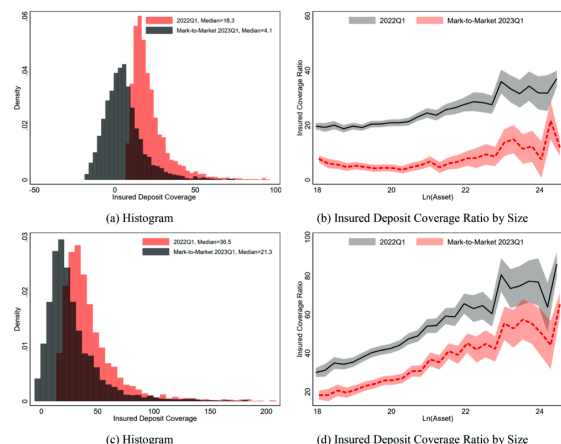
5.4 Figure 4: Distribution of Insured Deposit Coverage Ratio under Different “Run” Cases + Table 2: Insolvent Banks Under Different Cases

Read:

- Negative insured deposit coverage means remaining marked-to-market assets cannot repay insured deposits after uninsured withdrawals.
- Under all uninsured depositors running, 1,619 banks with \$4.9 trillion in assets have negative insured-deposit coverage.
- Under 50% uninsured withdrawal, 186 banks with about \$0.3 trillion in assets remain vulnerable.
- Under the all-assets-liquidate benchmark, 2,315 banks with \$11 trillion in assets have negative capitalization relative to non-equity liabilities.

Detailed interpretation: This figure/table pair is an FDIC-perspective stress test. It quantifies both the number of vulnerable banks and the scale of assets and shortfalls associated with them.

Economic message: Uninsured runs can impair insured deposits and therefore create a policy-relevant externality.



of Insured Deposit Coverage Ratio under Different “Run” Cases.

stograms (density) of the insured deposit coverage ratio calculated based on Q1 2022 balance sheets and mark-to-market values, as well as these values across bank size (Panels b and d). We simulate two cases. In the first case (Panels a and c), we assume that insured deposits are fully covered. In the second case (Panels b and d), we assume half of insured deposits are covered.

(a) Figure 4: Insured Deposit Coverage Ratio under Different Run Cases

Table 2
Insolvent Banks Under Different Cases.

	(1) All Assets Liquidate	(2) 100 % Uninsured Depositor Run	(3) 50 % Uninsured Depositor Run
Aggregate Asset	11T	4.9T	0.3T
Aggregate Equity	1.0T	0.4T	0.02T
Aggregate Insured Deposit	5.2T	2.6T	0.25T
GIB Banks	3.5T	1.5T	0.28T
Total Shortfall	1.3T	0.3T	0.01T
GIB Banks	1T	0.2T	0.04T
Total Asset	0.48	0.38	0.28
Liability/Asset	(68)	(48)	(98)
Domestic Deposit/Asset	91.7	91.9	92.0
Insured Deposit/Asset	(2.3)	(2.3)	(3.6)
Domestic Deposit/Asset	89.6	90.7	90.8
Insured Deposit/Asset	(4.9)	(3.1)	(3.7)
Domestic Deposit/Asset	66.4	67.8	79.7
Insured Deposit/Asset	(11.6)	(11.4)	(5.8)
Domestic Deposit/Asset	22.1	22.4	10.2
Insured Deposit/Asset	(11.7)	(11.8)	(7.2)

(b) Table 2: Insolvent Banks Under Different Cases

5.5 Figure 5: Full Set of Insolvent Banks if All Uninsured Depositors Run (Insured Deposit Coverage Ratio Metric)

Read:

- The y-axis is mark-to-market asset losses as a percentage of initial assets.
- The x-axis is uninsured deposits as a percentage of marked-to-market assets.
- SVB sits in the top-right region: large asset losses and very high runnable uninsured deposits.
- Red dots mark the ten largest vulnerable banks, including one above \$1 trillion in assets.

Detailed interpretation: This scatterplot is the clearest visual expression of the mechanism: neither asset losses nor uninsured leverage alone is enough; the dangerous banks have both.

Economic message: Run vulnerability is a joint balance-sheet event.

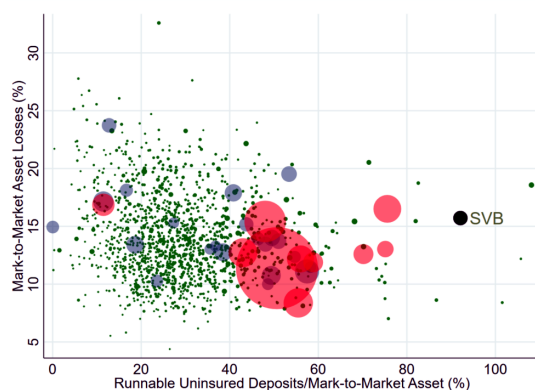


Figure 5: Full Set of Insolvent Banks if All Uninsured Depositors Run

5.6 Figure 6: Insolvent Banks under Different Uninsured Deposits Runs Cases (Insured Deposit Coverage Ratio Metric)

Read:

- The number and aggregate assets of insolvent banks rise with the share of uninsured deposits withdrawn.
- At 10% withdrawals, 66 banks with about \$210 billion of assets are vulnerable.
- At 30% withdrawals, 106 banks with about \$250 billion of assets are vulnerable.
- At 100% withdrawals, 1,619 banks with about \$4.9 trillion of assets are vulnerable.

Detailed interpretation: The figure provides a sensitivity curve rather than a single stress test. It shows how latent fragility grows as more depositors become awake.

Economic message: Even partial uninsured withdrawals can generate failures under the insured-deposit-coverage metric.

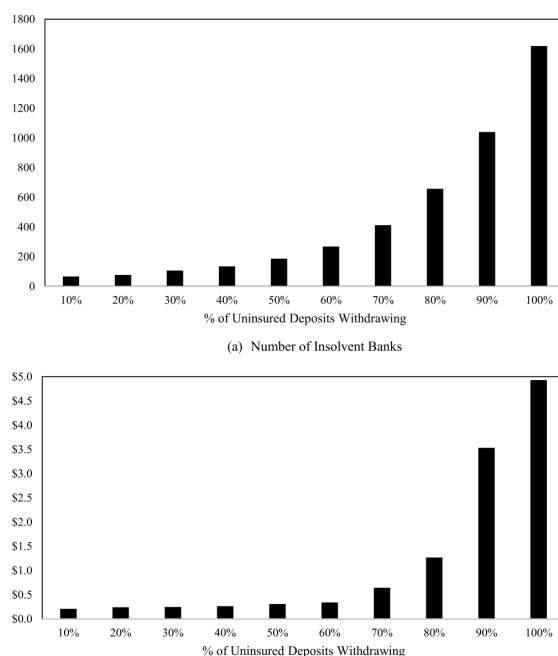


Figure 6: Insolvent Banks under Different Uninsured Deposit Run Cases – Insured Deposit Coverage Metric

5.7 Figure 7: Insolvent Banks under Different Uninsured Deposits Runs Cases (Capital Ratio Metric)

Read:

- The number of negative-capital banks rises with uninsured withdrawal shares.
- Aggregate assets of affected banks can reach about \$1.4 trillion.
- SVB would fail at a 50% uninsured withdrawal threshold under this metric; First Republic at 70%.
- The capital-ratio metric generally produces fewer vulnerable banks than the insured-deposit-coverage metric.

Detailed interpretation: This metric focuses on book-equity impairment after forced asset sales. It is narrower than insured-deposit coverage, but it still finds economically important fragility.

Economic message: The conclusion is robust across metrics: monetary tightening created broad latent bank fragility.

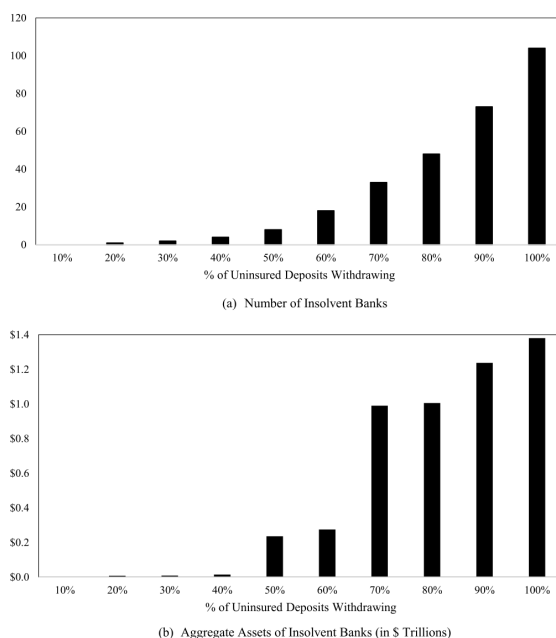


Figure 7: Insolvent Banks under Different Uninsured Deposit Run Cases – Capital Ratio Metric

6 Short summary

- Aggregate U.S. bank asset values fell by roughly \$2 trillion relative to book value.
- SVB's distinctive feature was extreme uninsured leverage, not uniquely severe asset losses.
- The model shows that even liquid banks can face self-fulfilling solvency runs.
- Under all-uninsured withdrawal, 1,619 banks with \$4.9 trillion in assets have negative insured-deposit coverage.
- Under a 50% uninsured run, 186 banks with about \$0.3 trillion in assets remain vulnerable.
- A credible stress test must jointly measure asset duration losses, capital, deposit franchise, and uninsured leverage.

7 Conclusion

7.1 Core conclusion

The 2022–2023 monetary tightening produced large hidden mark-to-market losses in U.S. banks, and those losses interacted with uninsured leverage to create latent solvency-run risk.

7.2 Economic intuition

Higher rates reduce long-duration bank assets. If uninsured depositors run, the bank loses franchise value exactly when it needs it, and selling assets at market values can make the bank insolvent.

7.3 Takeaway

Bank fragility cannot be evaluated from book capital alone. A credible stress test must jointly measure duration losses, capital, deposit franchise, and uninsured leverage.